Project Scope

Note: This document is intentionally incomplete and is in non-standard format; as part of the course this document will be re-written.

Project Scope: (ScrumFall process)

Saving a life is a noble cause. But natural disaster events like floods, earthquake, tornado, tsunami and hurricane occur too often and too suddenly claiming very many lives. Human and animal lives along with their assets are in danger. There is a need for a global reach through Virtual Disaster Assistance.

Request for Help

People who are exposed to such nature's fury need help. Help is required in the form of human rescuers, money and materials. Various categories of items like, food, clothing shelter, medicines, medical attention, and common day to day consumables are required. Need for volunteers who can be on the field, is very critical. Transport on land, water and air is required. Such events needing help may occur simultaneously at multiple locations. **Optimized use of available resources is a critical success factor.** Anyone can **request help.** A request for Help may have one or more components (Human resources, money, material) in it. The requestor or the relief center person can **expire** individual request items or the entire request. If a request has been fulfilled it is so marked. Both fulfilled and expired items can't be delivered against. There are many categories of items and categories of help that may be required, and it does become a challenge to collect, process, deliver and manage such requests.

Response to Help

Fortunately, there are very many good Samaritans who are keen to help but they may not know what is required and where such help needs to be delivered. Both coordination and automation are required for the success of the relief and rescue efforts. Anyone can **respond** to a request. The response may be towards one or more items in a request or all of it. A response will be to one and only one relief center. A response to help may be proactively (even before a disaster has occurred) made but delivered against a real event. **This is known as a pledge.** We certainly need a call center that connects Donors, Recipients, the volunteers and government agencies. **But our phase 1 goal is to have the request-response matched automatically as much as possible using the proximity of locations, to have an automated relief management.** The users should be able to query and match the request that they wish to respond and deliver to the specific location directly. There is No storage capacity with this system.

The data model should be designed in such a way that users can themselves search for the requirements at relief center near their Zip code. It is considered most efficient if people near the location of disaster provide the help. The proposed web portal must be highly extensible and support concurrency. The goal of this phase of project is to create a robust data model and design the database using the principles of our course. And to develop a web application interface to as many functions as possible.

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Some Objectives and capabilities of this tool are as follows:

a. Call center users can connect those who need help, those who wish to help, volunteers and government agencies. Call center operator and admin users.
b. Disaster events with geographic location details, its contact center details, duration of occurrence of event and the categories of required items or help can be created by admins.
c. Once an event is created individual items required at each location can be entered by any user from the event center or administrator
d. The categories of items will differ from event to event.
e. Assume common attributes of each item to be listed.
f. Confidentiality of volunteers must be maintained.
g. Email Confirmation is required.
h. Context sensitive help is required, and the help text should be changeable without changing software
i. Capability to Change or Cancel a Requirement or the offer to help is required.
j. System Administrators can do any of the functions
k. User Feedback, ranking on condition of resource can be maintained

An urgent requirement is to build a state-of-the-art web and mobile application for such a tool. An iterative approach is probably ideal. Recommend using Scrum and WaterFall process hybrid.

While we have a good idea of the requirements, we would like help from your group to draft the requirements and develop this application within the next 3 months. Anticipate lots of changes. The broad scope is

1) **Donor functions: Respond, Pledge to respond, etc.**
2) **Disaster Event center representative (Recipient) functions include register for help, receive help, expire help etc.**
3) **Administrator functions must be provided.**
4) **All users will require to be authenticated. The tool must ensure privacy of users and security of data.**
5) **Email notifications**

The key is to come up with a good architecture which can support both Mobile and Web browser environments.

Code needs to be layered and completely reused between the two. Initial version can support just the web browser environment.

Such software is expected to attract worldwide users. It must be simple software, easy to use, maintain and offer very high performance. While we have a good idea of the requirements, we would like help from your group to draft specific requirements and develop this application within the next 3 months. Customer contacts will provide additional details of requirements, if requested by the Project team.

Project Scope

Disaster Relief Management System (DRMS)

Customer contacts: Raman, Karan

General Requirements:

2. **Must use the principles of Software Engineering**
3. **Must use Enterprise Architect tool.**
4. **Use tools is recommended. This environment setup should be the zeroth iteration of the project. The tools may be used as a service or installed in your local. Note ITS and CSG offer many options.**
    a. **source library control (github etc.)**
    b. **automated tests (jUnit, Cucumber,selenium etc.)**
    c. **automated Continual code inspection (sonar etc.)** https://www.sonarqube.org/
    d. **Continual integration (CI) (Cruise control, Teamcity, Gitlab)** http://cruisecontrol.sourceforge.net/
    e. **Defect tracking (Bugzilla etc.)** https://www.bugzilla.org/
5. **The tool should be user friendly & require minimal data entry. The Software development team and the Business analysts are required to suggest the specific user friendly features while drafting the User Requirement Specifications. Some examples of user friendly features may be**
    a. **minimum data entry**
    b. **data transfer from context sensitive help screens**
6. **The system must support user Id / Password based authentication and must offer the best security features. The software development team is required to identify (in URS) the specific security features to be incorporated in VSS. Some examples of such security features may be**
    a. Data encryption techniques (128 bit)
    b. Invisible password
    c. Structure of Password
    d. Password change policy
    e. User Id / Password retrieval policy
7. **The System must provide four categories of users with role-based access control (RBAC). Should be able to add more.**
    a. Donors
    b. Event Center representative (Recipient)
    c. Call center operators
    d. System Administrators
8. **Objectives and capabilities of this tool are as follows:**
    a. Authentication capabilities must be cutting edge.
    b. Sys Administration tasks
    c. Additional use cases are being defined.
    d. Donor and Recipient functions.

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Non-Functional, Operational requirements from project team:

- **Mysql or Oracle or Sql Server database**
- **Backend code to be developed using Java/.Net/Python**
- **Use of MVC / REST or one of the current architectures**
- **UI must be responsive and 508 compliant**
- **Each Team must pick a Team Leader and must rotate the roles.**
- **Teamwork is very important**
- **Team members must have a minimum of 1-2 meetings a week details of which must be recorded.**
- **Meetings should be well planned and should preferably be less than 30 minutes in each meeting.**
- **It is highly recommended that teams meet once right after class every week.**
- **Distributed teams are a matter of fact; technology should be used to meet from remote, if it is convenient.**
- **Team must submit more detailed schedule and get it approved.**
- **The team may choose programming tools get it approved.**
- **Use of tools and automation will be recognized. Automated testing is mandatory.**
- **The work done in each stage of software development must be acceptable to the customer before the team may proceed with the next stages.**
- **All members of the team must assume specific responsibilities for each activity of software development**
- **The customer must be informed of the Roles and Responsibilities**
- **The tools required for the project will NOT be provided by the customer**
- **Due to acute shortage of resources, the customer is able to meet with project teams only on a pre-arranged basis.**
- **All documents must be delivered using ICON dropbox.**
- **Teams must develop the Screens and demo the screen flow before developing the full version.**

Tools: The project team may select & use any tools required to deliver the functionality; but must require Customer approval.

Deliverables:

- All documents must be delivered using ICON drop box.
- Teams must develop the Screens and demo the screen flow before developing the full software.
- Teams need to submit an Iteration plan identifying the functions to be delivered.

Project Scope

Define-Analyze-Design-Develop-Test-Implement

Specific Deliverable schedule: ScrumFall

#	Deliverable Description	Due on or before
Plan	**Project Plan:** Documentation template, Master Project Schedule, Estimation work sheet, Roles and Responsibilities, Risk Management Plan, Tool plan **5%**	For each iteration
Define	**User Requirement Specification (URS) – Use cases 10%**	For each iteration
Test	**Test Plans, Automated Testing 10%**	For each iteration
Analyze	**Software Requirements Specification (SRS) Models, Analysis, Data Dictionary, Specific Requirements. 10%**	For each iteration
Design	**High Level Design(screens, database, architecture) 10%**	For each iteration
Develop, Test, Implement	**Iteration Zero – Setup Environment 5% See #4 in general requirements**	**Feb 15th**
	Working software Iteration #1 10%	**Mar 1st week**
	Working software Iteration #2 10%	**Apr 1st week**
	Working software Iteration #3 – Final 20%	**May 1st week**
Present	**Final Team Presentations 5%**	Last week of class
Reports	**Weekly Progress Reports 5%**	Every Tuesday

Revision Log

Date	Version	Description of change
1/28/2020	**1.0**	**Initial version**

Distribution List {Students, TA}